



RV College of Engineering®
(An autonomous institution affiliated to VTU, Belagavi)

Department of Master of Computer Applications

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IV SEMESTER – MCA491P Major Project
Context-Aware Academic Assistant Using Multi-Agent AI
And Retrieval Augmented Systems

Recent advancements in Artificial Intelligence, particularly in Natural Language Processing, have led to the development of Retrieval-Augmented Generation (RAG) systems that combine information retrieval with language models for generating context-aware responses. These systems are widely used in chatbots and knowledge assistants but remain static, as they do not learn from feedback or improve over time, leading to inaccurate or less reliable outputs. With the emergence of Agentic AI, there is an opportunity to design systems that can evaluate and refine their own responses. This project aims to develop a self-improving multi-agent RAG system that enhances response quality through continuous feedback and agent collaboration.

The system will follow a modular approach using technologies such as LangChain, LangGraph, Azure OpenAI, HuggingFace embeddings, and FAISS. The main modules include a Document Ingestion Module, Retriever Agent for fetching relevant data, Generator Agent for response creation, Critic Agent for evaluating outputs, and Improver Agent for refining responses. A Feedback Module will store user inputs and system evaluations to enable continuous learning. Streamlit will be used to develop a simple user interface. These tools are selected for their efficiency in building scalable, agent-based AI systems.

The expected outcome is an intelligent system capable of generating accurate and context-aware responses while improving over time. The system is anticipated to achieve better performance than traditional RAG systems, with an estimated 20–30% improvement in response quality through iterative refinement. This project demonstrates the application of multi-agent collaboration and feedback-driven learning, with potential use cases in education, legal systems, and customer support, while also contributing to research in adaptive AI systems.

Signatures

Student

Internal Guide

**Committee Member/
External Evaluator**

Director